

Experiment 32

Aim

To perform morphological closing on an image using OpenCV.

Procedure

1. Import the required libraries, OpenCV and NumPy.
2. Read the input image.
3. Convert the image into grayscale.
4. Create a morphological kernel.
5. Apply the Closing operation using cv2.morphologyEx().
6. Display the original and closed images.
7. Save the resulting image.

Code:

```
import cv2
import numpy as np
img = cv2.imread("image.jpg")
if img is None:
    print("Error: Image not found!")
    exit()
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
kernel = np.ones((5, 5), np.uint8)
closing = cv2.morphologyEx(gray, cv2.MORPH_CLOSE, kernel)
cv2.imshow("Original Image", img)
cv2.imshow("Closing Image", closing)
cv2.imwrite("closing_image.jpg", closing)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Sample Input:



Output:



Result

The morphological closing operation was successfully performed using OpenCV. The closing operation filled small holes and gaps in the objects of the image.